

Exp no:9

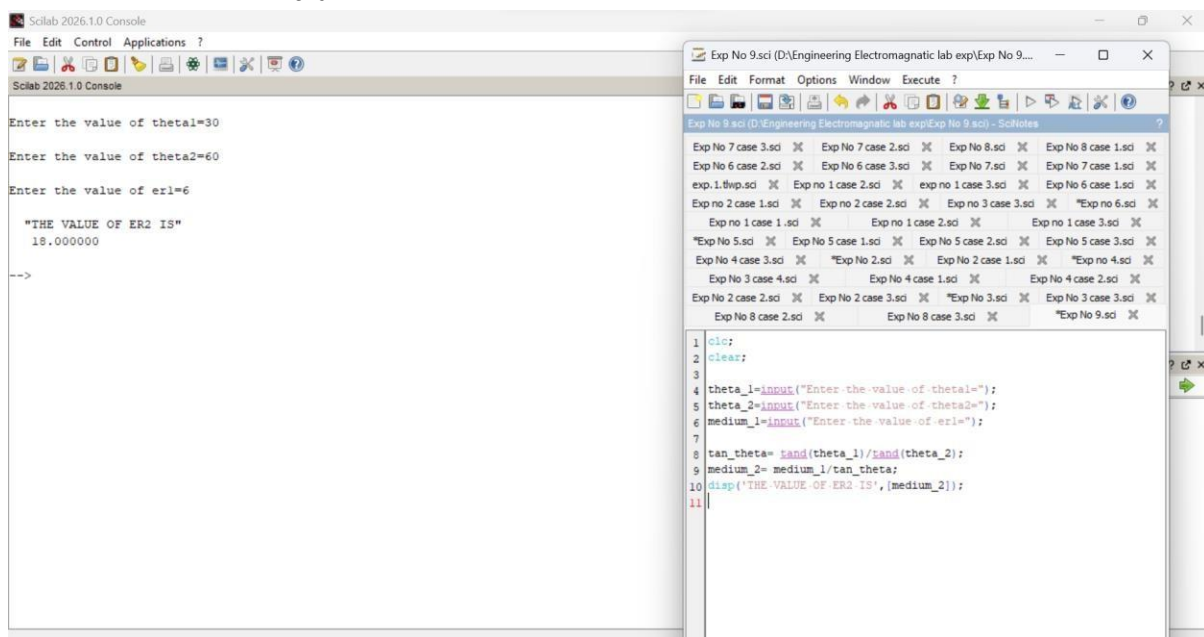
Implement Dielectric free space boundary conditions in Electric Field and find the relative permittivity of the different medium.

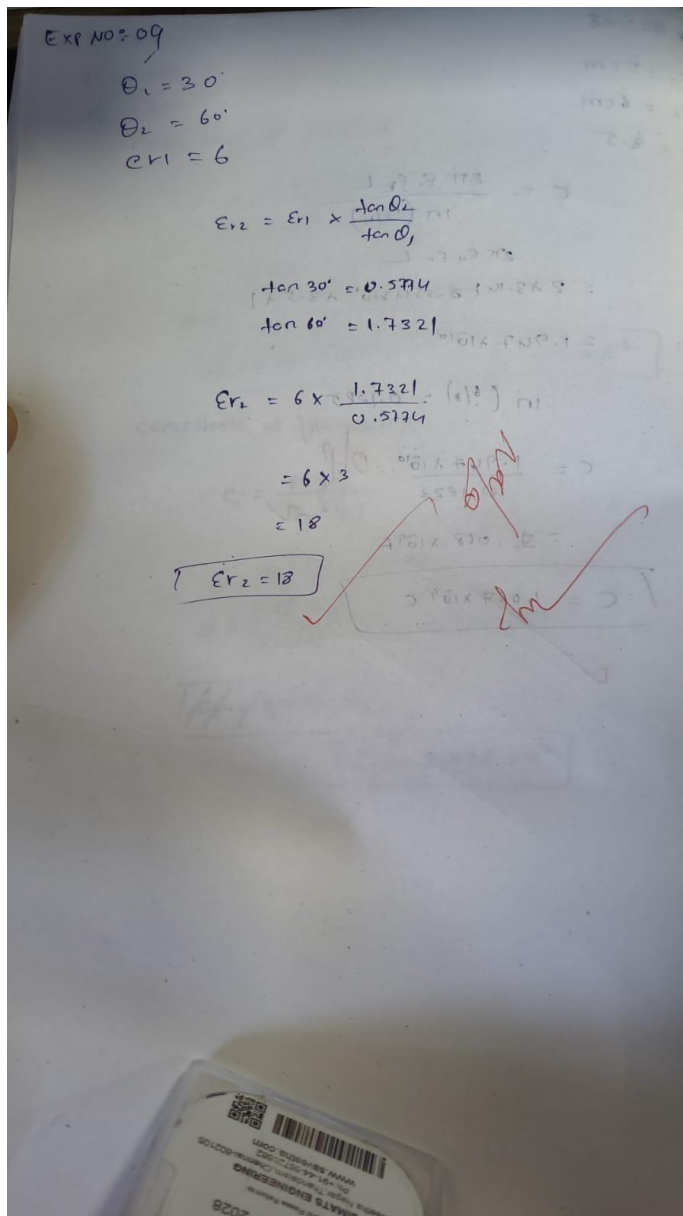
Aim:

To write program for Dielectric - free space boundary condition in Electric field using SCILAB.

Software required:

- SCILAB version 6.0.1





Case 1: In this case, the dielectric medium has a relative permittivity (ϵ_r) of 2, meaning the dielectric reduces the electric field by half compared to the free space. This is typical for materials like mica or certain types of plastics.

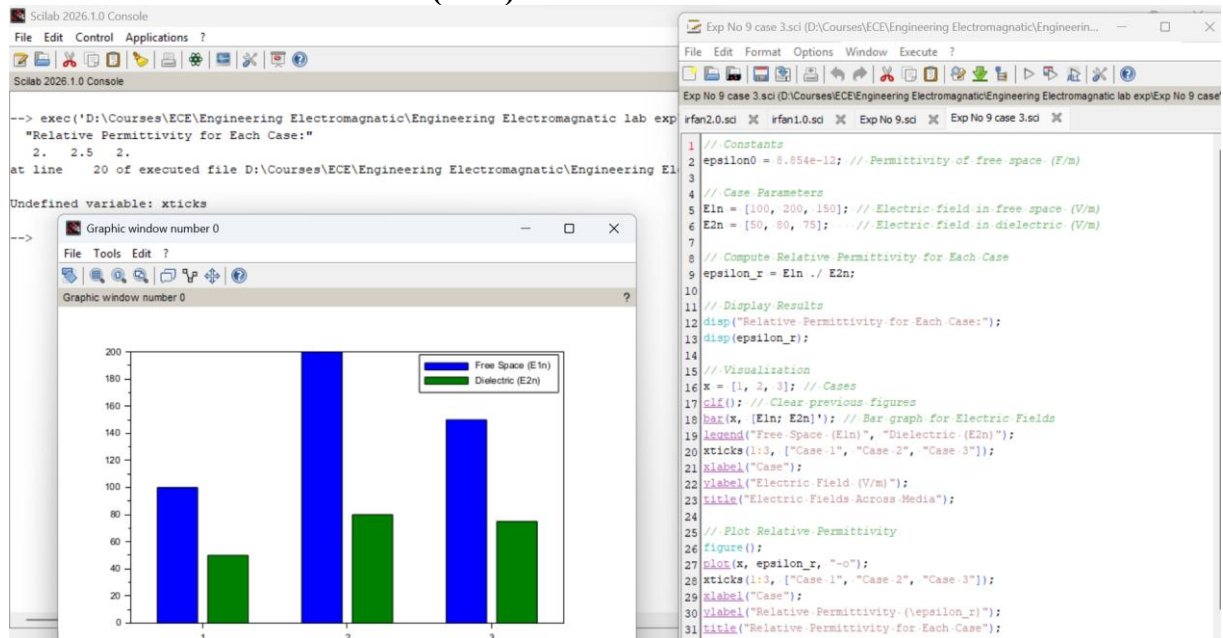
- **Free Space Electric Field (E_1):** 100 V/m
- **Dielectric Electric Field (E_2):** 50 V/m

Case 2: Here, the dielectric medium has a higher relative permittivity ($\epsilon_r=2.5$) compared to Case 1. This indicates a stronger polarization effect in the dielectric, as seen in materials like glass or water.

- **Parameters:**
 - $E_{n1}=200$ V/m (Electric field in free space).
 - $E_{2n}=80$ V/m (Electric field in the dielectric medium).

Case 3: The dielectric medium here also has a relative permittivity of 2. The behaviour of the material suggests it could be another low-polarization dielectric material, with a consistent reduction of the electric field by half.

- **Free Space Electric Field (E_{1n}):** 150 V/m
- **Dielectric Electric Field (E_{2n}):** 75 V/m



Result :

Thus the program was verified for Dielectric - free space boundary condition in Electric field using SCILAB was successfully.